11

INTRODUCING CSS

Makeup Class

- Decide the Time with Class

Style Separate from Structure

These pages have the exact same HTML source but different style sheets:

(csszengarden.com)

How Style Sheets Work

1. CSS stands for Cascading Style Sheets

2. CSS describes how HTML elements are to be displayed on screen, paper, or in other media

3. Attach the styles to the document (there are a number of ways).

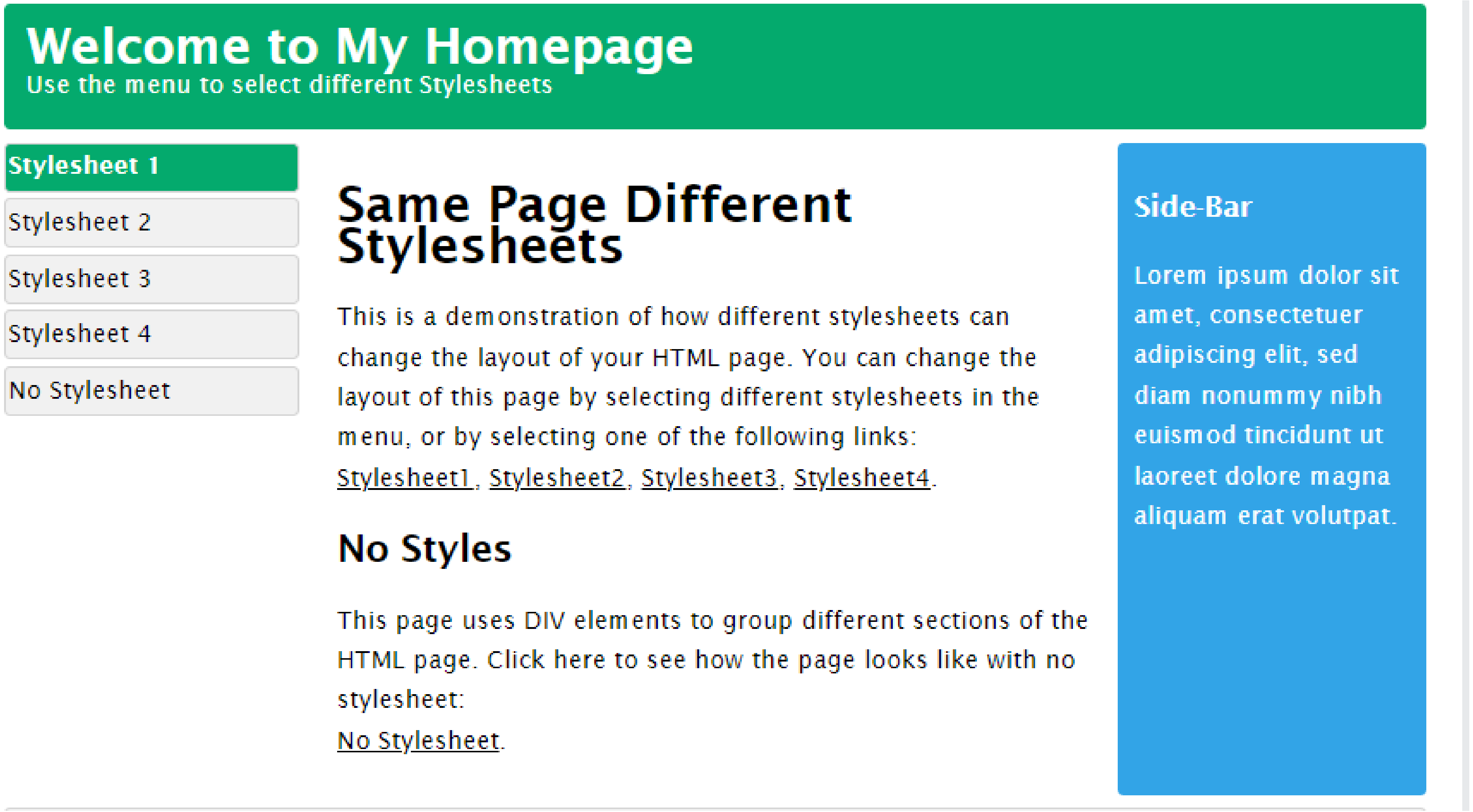
4. The browser uses your instructions when rendering the elements.

5. CSS saves a lot of work. It can control the layout of multiple web pages all at once

How Style Sheets Work

Here you will see one HTML page displayed with four different stylesheets.

Style 1



How Style Sheets Work

Here you will see one HTML page displayed with four different stylesheets.

Style 2

How Style Sheets Work

Here you will see one HTML page displayed with four different stylesheets.

Style 3

Welcome to My Homepage

Use the menu to select different Stylesheets

Stylesheet 1 | Stylesheet 2 | Stylesheet 3 | Stylesheet 4 | No Stylesheet

Same Page Different Stylesheets

This is a demonstration of how different stylesheets can change the layout of your HTML page. You can change the layout of this page by selecting different stylesheets in the menu, or by selecting one of the following links:
Stylesheet1, Stylesheet2, Stylesheet3, Stylesheet4.

No Styles

This page uses DIV elements to group different sections of the HTML page. Click here to see how the page looks like with no stylesheet:
No Stylesheet.

How Style Sheets Work

Here you will see one HTML page displayed with four different stylesheets.

Style 4

Types of CSS

CSS comes in three types:

- Internal CSS
- External CSS
- Inline CSS

Internal CSS

An internal CSS is used to define a style for a single HTML page.

Internal CSS can be applied by using a <style> element in the <head> section

It is also known as Embedded Style Sheets

For example:

If you want to set the text color of all<h1> value to blue, and the color of all <p> elements to red, and the background color to grey using Internal CSS then how can you do that?

Internal CSS

For example: If I want to set the text color of all<h1> value to blue, and the color of all <p> elements to red, and the background color to grey then:

```
<!DOCTYPE html>
<html>
<head>
<style>
body {background-color: lightgray;}
h1   {color: blue;}
p    {color: red;}
</style>
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

External CSS

An external style sheet is used to define the style for many HTML pages.

To use an external style sheet, add a link to it in the <head> section of each html page.

A separate, text-only .css file associated with the document with the link element or @import rule

For example:

If you want to set the text color of all<h1> value to blue, and the color of all <p> elements to red, and the background color to powderblue using external CSS then how can you do that?

External CSS

For example: If I want to set the text color of all<h1> value to blue, and the color of all <p> elements to red, and the background color to powderblue using external CSS then:

```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="styles.css">
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

"styles.css":

```
body {
  background-color: powderblue;
}
h1 {
  color: blue;
}
p {
  color: red;
}
```

External Style Sheets

The style rules are saved in a separate text-only *.css* file and attached via **`link`** or **`@import`**.

Via **`link`** element in HTML:

```
<head>
  <title>Titles are require</title>
  <link rel="stylesheet" href="/path/example.css">
</head>
```

Via **`@import`** rule in a style sheet:

```
<head>
  <title>Titles are required</title>
  <style>
    @import url("/path/example.css");
    p {font-face: Verdana;}
  </style>
</head>
```

Inline CSS

An inline CSS is used to apply a unique style to a single HTML element.

An inline CSS uses the <style> attribute of an HTML element.

For example:

If you want to set the text color of all<h1> value to blue, and the color of all <p> elements to red then how can you do that using inline CSS?

Inline CSS

For example: If I want to set the text color of all<h1> value to blue, and the color of all <p> elements to red then :

```
<h1 style="color:blue;">A Blue Heading</h1>

<p style="color:red;">A red paragraph.</p>
```

Inline Styles

Apply a style declaration to a single element with the **`style`** *attribute*:

```
<p style="font-size: large;">Paragraph text...</p>
```

To add multiple properties, separate them with semicolons:

```
<h3 style="color: red; margin-top: 30px;">Intro</h3>
```

Style Rules

Each rule *selects* an element and *declares* how it should display.

```
h1 { color: green; }
```

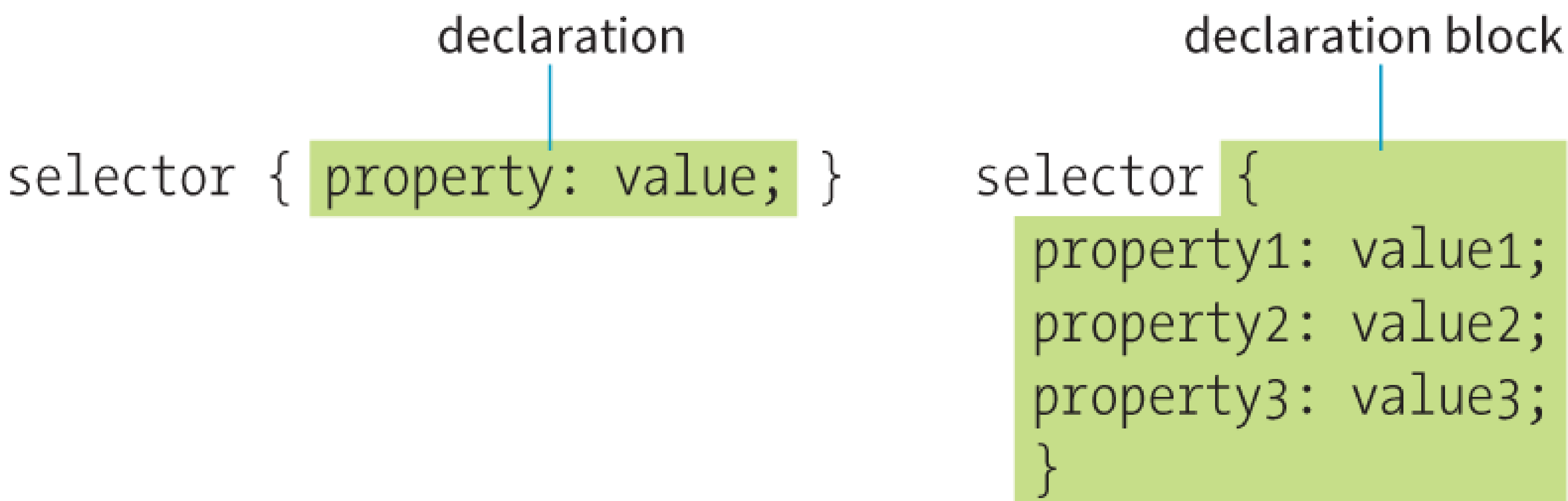
This rule selects all `h1` elements and declares that they should be green.

```
strong { color: red; font-style: italic; }
```

This rule selects all `strong` inline elements and declares that they should be red and in an italic font.

Style Rule Structure

- A style rule is made up of a **selector** a **declaration**.
- The declaration is one or more **property / value** pairs.



Selectors

There are many types of selectors. Here are just two examples:

```
p {property: value;}
```

Element type selector: Selects all elements of this type (**p**) in the document.

```
#intro {property: value}
```

ID selector (indicated by the # symbol) selects by ID value. In the example, an element with an id of “intro” would be selected.

Declarations

The **declaration** is made up of a **property/value pair** contained in curly brackets { }:

```
selector { property: value; }
```

Example

```
h2 { color: red;
     font-size: 2em;
     margin-left: 30px;
     opacity: .5;
   }
```

Declarations (cont'd)

- End each declaration with a semicolon to keep it separate from the next declaration.
- White space is ignored, so you can stack declarations to make them easier to read.
- **Properties** are defined in the CSS specifications.
- **Values** are dependent on the type of property:
  - Measurements
  - Keywords
  - Color values
  - More

CSS Comments

```
/* comment goes here */
```

- Content between `/*` and `*/` will be ignored by the browser.
- Useful for leaving notes or section labels in the style sheet.
- Can be used within rules to temporarily hide style declarations in the design process.

End of Lecture